Fahimul Haque

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SUMMARY

- Industrial and Production Engineering undergraduate at BUET (CGPA 3.44/4.00, after 7 semester) with 3 peer-reviewed journal and 5 conference papers.
- Research experience in statistical analysis, time-series forecasting, interpretable machine learning, and evolutionary game modeling; hands-on projects in manufacturing and product design.
- Seeking PhD opportunities in industrial engineering systems, with interests in manufacturing quality, operations and supply chain analytics, and socio-technical decision models.

EDUCATION

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh
B.Sc. in Industrial and Production Engineering (IPE) | CGPA: 3.44/4.00 (after 7 semester) Jan 2022 - Present

RELEVANT COURSEWORK

Operations Research, Operations Management, Supply Chain Management, Numerical Analysis, Product Design, Probability and Statistics.

RESEARCH INTERESTS

Manufacturing quality and defects, operations and supply chain analytics, optimization and decision modeling, interpretable machine learning, public health systems analytics, game theory.

PUBLICATIONS

- **Haque, F., Farid, S. B. (2025).** The Impact of GDP Growth on Infant Mortality Reduction: Statistical Analysis Over 20 Years in 30 Countries. *American Journal of Interdisciplinary Research and Innovation,* 4(2). DOI: 10.54536/ajiri.v4i2.4147.
- **Siddique, M. H., Haque, F. (2025).** A Multi-Agent Reinforcement Learning Approach for Recovering Evolutionarily Stable Strategies in Evolutionary Games Using Proximal Policy Optimization. *Proceedings of the 8th IEOM Bangladesh International Conference on Industrial Engineering and Operations Management*, Dhaka, Bangladesh, 20–21 December 2025. DOI: 10.46254/BA08.20250076
- **Haque, F., *Siddique, M. H.*, Zisan, M. R., Mahbub, N. (2025).** A Machine Learning–Driven Framework for Modeling and Forecasting Suicide Mortality in Asia: The Role of Multidimensional Socio-Economic Determinants. *Proceedings of the 8th IEOM Bangladesh International Conference on Industrial Engineering and Operations Management*, Dhaka, Bangladesh, 20–21 December 2025. DOI: 10.46254/BA08.20250013
- **Haque, F., Farid, S. B. (2025).** Government Healthcare Spending and Cancer Mortality: A Statistical Analysis Over 15 Years in 18 Countries. *Journal of Innovative Research,* 3(2). DOI: 10.54536/jir.v3i2.4201.
- **Haque, F. (2025).** Emission Trajectories in the Trade Context: A Comprehensive Machine Learning Approach Using K-Means and ARIMA. *Journal of Sustainable Engineering & Renewable Energy,* 1(1). DOI: 10.54536/jsere.v1i1.4747
- **Haque, F., Siddique, M. H., Zisan, M. R. (2025).** Interpretable Modeling of Flank Wear in Drilling Through Symbolic Regression and Ensemble Machine Learning Techniques. *Proceedings of the 8th International Conference on Mechanical Engineering and Renewable Energy 2025 (ICMERE2025)*, November 2025.
- **Haque, F., Zisan, M. R., Siddique, M. H. (2025).** Multi-Paradigm Predictive Modeling of Surface Roughness in Thermally Enhanced Friction Drilling of A356 Aluminum Alloy Using RSM–ML Integration. *Proceedings of the 8th International Conference on Mechanical Engineering and Renewable Energy 2025 (ICMERE2025)*, Chattogram, Bangladesh, November 2025.
- **Siddique, M. H., *Zisan, M. R.*, Haque, F. (2025).** Initialization-Free Non-Linear Constrained Optimization Using a Bayesian Self-Supervised MLP. *Proceedings of the 8th IEOM Bangladesh International Conference on Industrial Engineering and Operations Management*, Dhaka, Bangladesh, 20–21 December 2025. 10.46254/BA08.20250076

RESEARCH PROJECTS

Evaluating Suicide Risk Factors in Asia and Europe Using Machine Learning Dec 2023 - Ongoing

- *Supervisor: Dr. Ridwan Al Aziz (Assistant Professor, IPE, BUET).*
- World Bank indicators (2000-2021); PLS/OLS, bootstrapping, and ML models; interpretability via feature importance, PDP, and SHAP.

A Multi-Stakeholder Game-Theoretic Framework for Strategic Modeling of Digital Advertising Ecosystem Sep 2024 - Ongoing

- *Supervisors: Dr. Jun Zhuang (ISE, University at Buffalo), Dr. Puneet Agarwal (IME, Cal Poly), Dr. Ridwan Al Aziz (IPE, BUET).*
- Tripartite evolutionary game; replicator dynamics, simulations, and sensitivity analysis to study incentive design and misinformation funding.

POSTERS AND PRESENTATIONS

- Enhancing Fertility Rate Projections in South Korea Using Multivariate LSTM: Benchmark Against Traditional Models. CUET IEEE poster competition (2025). DOI: 10.13140/RG.2.2.35739.99362.

PROJECTS

Foundry and Metal Casting (IPE coursework) Jan 2023 - May 2023

- Pattern making, sand mold preparation, molten metal handling, and post-casting finishing.

SolidWorks: Tower Crane (SolidWorks lab) Apr 2023 - Jul 2023

- Full-scale 3D assembly in SolidWorks; large-assembly constraints and detailed part modeling.

Automatic Hammering Machine: PowerStrike (Product Design Sessional) Sep 2024 - Jul 2025

- Supervisor: Dr. Kais Bin Zaman (Professor, IPE, BUET).
- QFD, functional decomposition, feasibility and costing; CAD/CAE in SolidWorks and ANSYS; survey.

Greenhouse Automation (Computer Integrated Manufacturing project) Mar 2025 - Jul 2025

- Supervisor: Dr. Nafis Ahmad (Professor, IPE, BUET).
- Sensor monitoring and relay control using Arduino for temperature, humidity, soil moisture, light intensity.

INDUSTRY EXPOSURE

- GPH Ispat Limited (Internship) Nov 2025 – Dec 2025
- BAT Bangladesh (Visit)
- BM Energy Limited - LPG distribution plant (Visit)
- A1 Polymer Limited (Visit)

TECHNICAL SKILLS

- Programming: Python, MATLAB.
- ML/Analytics: time-series forecasting, clustering, ensemble ML, explainability (SHAP, PDP).
- CAD/CAE: AutoCAD, SolidWorks, ANSYS, CATIA.
- Tools: MS Excel, Word, PowerPoint, Power BI.

LEADERSHIP AND ACTIVITIES

- 1st Place, Undergraduate Research Competition, IEOM International Conference 2025 · Paper: Multi-Agent RL for Recovering ESS using PPO.
- Class Representative, Department of IPE, Class of 120, BUET (Jan 2023 – Jan 2024).
- Batch Representative, Batch of 1300, BUET (Mar 2025 – Present).
- Competitions: Bangladesh Economics Summit (1st round, 2025); National Social Business Case Competition (prefinal, 2025). Hult Prize (BUET campus round, 2025).
- International competitions: Astronomy and Astrophysics (prefinal, 2024); Youth Math Challenge (prefinal, 2022).

CERTIFICATIONS

- IBM (online): Python, Statistics, SQL, Project Management (2024-2025).
- Vanderbilt University (online): MATLAB (2025).